



K.R. MANGALAM UNIVERSITY
THE COMPLETE WORLD OF EDUCATION



FROYO
TECHNOLOGIES
DATA AI FOR BUSINESS



Intellithon'26: Problem Statements

Problem Statement #1: Predatory Loan Clause Detector

The Challenge:

In India's rural microfinance ecosystem, borrowers frequently fall into debt traps due to **exploitative and non-transparent loan clauses** embedded within contracts. These agreements are often written in complex legal language or regional scripts, making them difficult to understand for the average borrower.

Existing AI solutions primarily focus on English corporate contracts and fail to address the needs of underserved populations, leaving a critical gap in financial protection and accessibility.

Objective:

Design and develop a **mobile-first AI-powered solution** that:

- Extracts text from contract images using OCR (supporting regional languages)
- Identifies potentially harmful or non-compliant clauses based on RBI guidelines
- Provides **clear, concise, and user-friendly explanations** of risks in simple language (2-line summaries)

Expected Outcome:

- End-to-end **OCR + NLP pipeline**
- Intelligent **risk detection and clause classification model**
- User-friendly mobile/web interface for easy interaction and understanding

Problem Statement #2: SmartDorm — AI-Based Energy Optimization System

The Challenge:

Student accommodations such as hostels and PGs suffer from **significant energy inefficiencies**, driven by behavioral patterns like air conditioners running in unoccupied rooms, geysers left on for extended periods, and lights not being switched off.

Additionally, wardens lack real-time visibility into energy consumption, while students have little incentive to conserve electricity due to **fixed rental pricing models**, resulting in excessive energy waste and increased operational costs.

Objective:

Design and implement an **AI-driven energy management system** that:

- Monitors and analyzes **room-level energy consumption** (real or simulated data)
- Detects anomalies and identifies energy wastage patterns using ML techniques
- Provides intuitive dashboards for both administrators and students
- Encourages responsible usage through **gamification and incentive mechanisms**

Expected Outcome:

- Interactive **dashboard with real-time insights**
- ML-based **anomaly detection system**
- Automated **alert and notification system**
- Gamified features such as **leaderboards and reward mechanisms**

Instructions for Participants

1. Each team must select **one problem statement only** for participation.
2. All submissions must be **original**. Plagiarism or copied solutions will lead to immediate disqualification.
3. Teams are required to build a **working prototype (minimum viable product)** based on the selected problem statement.
4. The use of modern technologies such as **AI/ML, Web Development, APIs, or IoT** is encouraged wherever relevant.
5. Teams must prepare a **presentation/demo** explaining:
 - Problem understanding
 - Proposed solution
 - Technical approach
 - Impact and scalability
6. All required submissions must be completed **within the given timeline**. Late submissions will not be considered.
7. Ensure proper **team coordination and communication** throughout the hackathon.
8. Follow all **event guidelines and code of conduct** during the hackathon.

Innovate with purpose, build with passion, and present with confidence!